SASHA RANA

San Luis Obispo, CA • Sammamish, WA
(425) 270-9149 • sasha.rana.la@gmail.com • LinkedIn: Sasha Rana

EDUCATION

B.S. Mechanical Engineering | California Polytechnic State University, San Luis Obispo Sep 2025 – Aug 2029
San Luis Obispo, CA | Concentration: Mechatronics | Pursuing Minor in Computer Science

ENGINEERING EXPERIENCE

Mechanical Design Engineer Intern | Ethosphere Jun 2026 – Aug 2026
Seattle, WA | Hybrid

- Designed wearable voice-capture hardware in SolidWorks, owning end-to-end device packaging and compact enclosure development.
- Packaged a dense PCB with integrated buttons, LED light pipes, magnetic mounting, and Class-A surfacing while refining wall thicknesses, tolerances, and fit.
- Developed mechanical designs for both rapid 3D-printed prototypes and production-ready hardware.

Chassis Engineer | Cal Poly Racing – Formula SAE Sep 2025 – Present
San Luis Obispo, CA

- Contribute to the design and fabrication of carbon-fiber monocoque chassis, firewall, and metal frontal-loop structure, with emphasis on driver safety and structural stability.
- Designed an anti-submarine harness mounting system with aluminum inserts, spreaders, and backing plates to withstand 15 kN attachment loads while meeting Formula SAE mounting constraints.
- Create and refine chassis CAD in Siemens NX for Formula SAE vehicle development.

Robotics Team Member | Gear Slingers – Cal Poly VEX U Robotics Sep 2025 – Present
San Luis Obispo, CA

- Build and assemble competition robots, modeling components and assemblies in Onshape for the VEX U Robotics Competition.
- Prototype and iterate an elevator-mounted cone intake, integrating an angled claw to securely grab cones and lift them through the full elevator travel.

TECHNICAL SKILLS

Engineering & Design: SolidWorks, Siemens NX, Onshape, mechanical design, design for manufacturing, GD&T fundamentals

Manufacturing & Fabrication: Manual mill, lathe, CNC machining, drill press, band saw, welding, 3D printing, precision measurement

Programming & Analysis: Python, C/C++, MATLAB, data analysis

Engineering Tools: CAD modeling, FEA fundamentals, engineering drawings, technical documentation

ADDITIONAL EXPERIENCE

Membership Sales Associate | 425 Fitness Mar 2025 – Sep 2025
Issaquah, WA

- Managed front-desk operations, answered member questions, led facility tours, and supported membership sales.

Camp Counselor / Full-Time Coach | Bellevue College Nike Soccer Camp Jun 2024 – Sep 2024
Bellevue, WA

- Led groups of up to 20 children ages 6–14 through soccer drills and games, building communication, leadership, and mentorship skills.